Chapter 1: Introduction to Computer Systems

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Operating systems manage hardware resources and provide abstractions for software.

Key concepts include virtualization, concurrency, and persistence.

Chapter 2: Virtual Memory and Paging

Virtual memory creates the illusion of a large, contiguous address space.

The CPU Memory Management Unit translates virtual addresses to physical frames.

Multi-level page tables save memory by allocating tables on demand.

Chapter 3: Concurrency and Multi-threading

A thread is a lightweight execution context within a process.

Mutual exclusion locks prevent race conditions on shared data.

Deadlocks occur when threads hold resources while waiting for others.